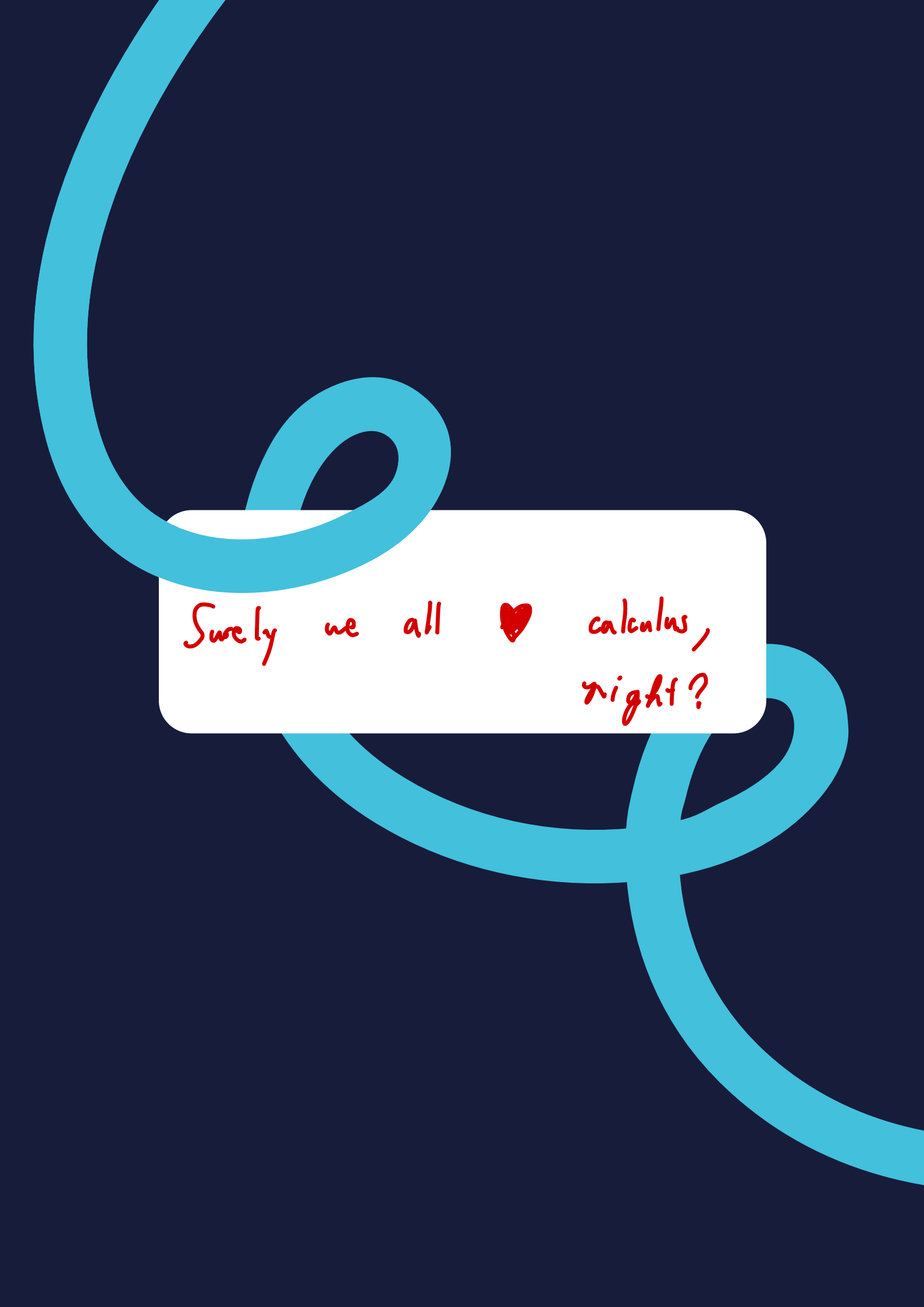




Surely we all ♥ calculus,  
right?

1.2 Composite function  $(f \circ g)(x) = f(g(x))$

1.5  $\lim_{x \rightarrow a} f(x) = L$  iff  $\lim_{x \rightarrow a^+} f(x) = L$  and  $\lim_{x \rightarrow a^-} f(x) = L$

1.6 Limit Laws, only true when  $\lim_{x \rightarrow a} f(x)$  exists  $= L$  and  $\lim_{x \rightarrow a} g(x)$  exists  $= M$

a.) Additivity Limit Law:  $\lim_{x \rightarrow a} [f(x) + g(x)] = L + M$

b.) Difference Limit Law:  $\lim_{x \rightarrow a} [f(x) - g(x)] = L - M$

c.) Constant Multiple Limit Law:  $\lim_{x \rightarrow a} [c f(x)] = c \cdot L$

d.) Product Limit Law:  $\lim_{x \rightarrow a} [f(x) g(x)] = L \cdot M$

e.) Quotient Limit Law:  $\frac{\lim_{x \rightarrow a} f(x)}{\lim_{x \rightarrow a} g(x)} = \frac{L}{M}$ , provided that  $M \neq 0$

f.) Power Limit Law:  $\lim_{x \rightarrow a} [f(x)]^n = L^n$ , where  $n$  is a positive integer

Special cases:

Constant Limit Law:  $\lim_{x \rightarrow a} c = c$ , where  $c$  is a constant

\*  $\lim_{x \rightarrow a} x = a$

\* g.) Polynomial Limit Law:  $\lim_{x \rightarrow c} P_n(x) = P_n(c)$ , where  $n \in \mathbb{Z}^+$ ,  $P_n(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0$

↳ Derived from:  $\lim_{x \rightarrow c} x^n = c^n$ , where  $n \in \mathbb{Z}^+$

\* h.) Root Limit Law:  $\lim_{x \rightarrow a} \sqrt[n]{f(x)} = \sqrt[n]{L}$ , where  $n \in \mathbb{Z}^+$ , if  $n$  is even,  $f(x) > 0$

↳ Derived from:  $\lim_{x \rightarrow a} \sqrt[n]{x} = \sqrt[n]{a}$ , where  $n \in \mathbb{Z}^+$ , if  $n$  is even,  $a > 0$

\* All parts of g.), h.) are also known as the direct substitution law/property.

↳ if  $f$  is a polynomial or rational function and  $a \in \mathcal{D}_f$ ,  $\lim_{x \rightarrow a} f(x) = f(a)$

Unknown Theorem 1: if  $f(x) \leq g(x)$  when  $x \rightarrow a$  (but  $a \neq 0$ ) and  $\lim_{x \rightarrow a} f(x)$  and  $\lim_{x \rightarrow a} g(x)$  both exist, then  $\lim_{x \rightarrow a} f(x) \leq \lim_{x \rightarrow a} g(x)$

Squeeze Theorem: If  $f(x) \leq g(x) \leq h(x)$  when  $x \rightarrow a$  (but  $a \neq 0$ ), and  $\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L$  then  $\lim_{x \rightarrow a} g(x) = L$

Composite Function Limit Theorem: If  $f$  is continuous at  $b$  and  $\lim_{x \rightarrow a} g(x) = b$ , then  $\lim_{x \rightarrow a} f(g(x)) = f(b)$ , i.e.  $\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x))$ , basically trafficking the limit symbol in and out.

## 1.7 Rigorous definition of Limits:

$\lim_{x \rightarrow a} f(x) = L$  implies that  $\forall \epsilon > 0, \exists \delta > 0$  s.t. if  $0 < |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon$

Rigorous definition of Left Hand Limits:

$\lim_{x \rightarrow a^-} f(x) = L$  implies that  $\forall \epsilon > 0, \exists \delta > 0$  s.t. if  $a - \delta < x < a \Rightarrow |f(x) - L| < \epsilon$

Rigorous definition of Right Hand Limits:

$\lim_{x \rightarrow a^+} f(x) = L$  implies that  $\forall \epsilon > 0, \exists \delta > 0$  s.t. if  $a < x < a + \delta \Rightarrow |f(x) - L| < \epsilon$

\* Rigorous definition of the infinite limits:

$\lim_{x \rightarrow a} f(x) = \infty$  implies that  $\forall M > 0, \exists \delta > 0$  s.t. if  $0 < |x - a| < \delta \Rightarrow f(x) > M$

\* Rigorous definition of -ve infinite limits:

$\lim_{x \rightarrow a} f(x) = -\infty$  implies that  $\forall N < 0, \exists \delta > 0$  s.t. if  $0 < |x - a| < \delta \Rightarrow f(x) < N$

\* There can be LH and RH limits that are +ve and -ve infinity, definition varies.

## 1.8 Continuity

Definition of continuity at a point:

$f$  is continuous at number  $c$  if  $\lim_{x \rightarrow c} f(x) = f(c)$

↳ This implies 1.  $f(c)$  is defined,  $a \in \mathcal{D}_f$

2.  $\lim_{x \rightarrow c} f(x)$  exists

3.  $\lim_{x \rightarrow c} f(x) = f(c)$

Definition of continuity from the right:

$f$  is continuous from the right at number  $c$  if  $\lim_{x \rightarrow c^+} f(x) = f(c)$

Definition of continuity from the left:

$f$  is continuous from the left at number  $c$  if  $\lim_{x \rightarrow c^-} f(x) = f(c)$

Continuity Theorems, if  $f$  and  $g$  are cont. at  $a$ , these are also cont. at  $a$ .

a.) Additivity Continuity Theorem:

$f+g$  is also cont.

b.) Difference Continuity Theorem.

$f-g$  is also cont. at  $a$

c.) Constant Multiple Continuity Theorem.

$cf$  is also cont. at  $a$ ,  $c$  is a constant

d.) Product Continuity Theorem.

$fg$  is also cont. at  $a$

e.) Quotient Continuity Theorem

$f/g$  is also cont. at  $a$ , if  $g(a) \neq 0$

f.) Polynomial Function Continuity Theorem

Any polynomial  $P_n(x)$  is cont. everywhere, i.e. cont. on interval  $(-\infty, \infty)$

g.) Rational Function Continuity Theorem

Any rational function is cont. whenever it is defined, i.e. cont. on its domain

h.) Trigonometric Function Continuity Theorem

Any trigonometric function is cont. at every number in their domain  
i.e.  $\sin x, \cos x$  cont. at interval  $(-\infty, \infty)$

$\tan x$  is **discont.** whenever  $x = n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$

$\csc x$  is **discont.** whenever  $x = n\pi, n \in \mathbb{Z}$

$\sec x$  is **discont.** whenever  $x = n\pi + \frac{\pi}{2}, n \in \mathbb{Z}$

i.) Root Function Continuity Theorem

Any root function  $\sqrt[n]{x}, n \in \mathbb{Z}^+$  is cont. at every number in their domain

j.) Composite Function Continuity Theorem.

If  $g$  is continuous at  $a$  and  $f$  is continuous at  $g(a)$ , then the composite function  
 $f \circ g(x) = f(g(x))$  is cont. at  $a$

★ Intermediate Value Theorem:

- Suppose that  $f$  is continuous on the closed interval  $[a, b]$  and let  $N$  be any number b/w  $f(a)$  and  $f(b)$ , where  $f(a) \neq f(b)$ , i.e.  $f(a) > N > f(b)$  or  $f(a) < N < f(b)$

- Then there exists a number  $c$  in  $(a, b)$  s.t.  $f(c) = N$

## Mathematical Induction for $n \in \mathbb{Z}^+$

- 1) Prove that  $S_1$  is true
- 2) Assume  $S_k$  is true, show that  $S_{k+1}$  is true
- 3)  $S_n$  is true  $\forall n \in \mathbb{Z}^+$

## 2.1 Derivatives

Definition of tangent line:

Slope of tangent line at point  $P(a, f(a))$  is:

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

Similarly, if  $h = x - a$ , when  $x \rightarrow a$ ,  $h \rightarrow 0$ , hence:

$$m = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h}$$

Definition of the derivative of a function  $f$  at a number  $a$

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(a+h) - f(a)}{h} \quad \text{or} \quad f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

## 2.2 Derivative as a function

Definition of the derivative of  $f$ :

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

Definition of differentiability:

$f$  is differentiable at  $a$  if  $f'(a)$  exists. It is differentiable on an open interval  $(a, b)$  [or  $(a, \infty)$  /  $(-\infty, a)$  /  $(-\infty, \infty)$ ] if it is differentiable at every number in the interval.

Diff. gives Cont. Theorem.

If  $f$  is differentiable at  $a$ ,  $f$  is continuous at  $a$ .

\*Note vice versa is false.

## 2.3 Differentiation Formula.

a.) Constant derivative theorem

$$\frac{d}{dx} c = 0, \text{ if } c \text{ is a constant}$$

b.) Power Rule, for any  $n$ .

$$\frac{d}{dx} x^n = nx^{n-1}$$

c.) Constant Multiple Rule

$$\frac{d}{dx} c f(x) = c \frac{d}{dx} f(x), \text{ if } c \text{ is a constant}$$

Following applies when  $f(x)$  and  $g(x)$  both exist,

d.) Additivity derivative rule

$$\frac{d}{dx} [f(x) + g(x)] = \frac{d}{dx} f(x) + \frac{d}{dx} g(x)$$

e.) Difference derivative rule

$$\frac{d}{dx} [f(x) - g(x)] = \frac{d}{dx} f(x) - \frac{d}{dx} g(x)$$

f.) Product rule.

$$\frac{d}{dx} [f(x)g(x)] = f(x) \frac{d}{dx} g(x) + g(x) \frac{d}{dx} f(x)$$

g.) Quotient rule

$$\frac{d}{dx} \left[ \frac{f(x)}{g(x)} \right] = \frac{g(x) \frac{d}{dx} f(x) - f(x) \frac{d}{dx} g(x)}{[g(x)]^2}, g(x) \neq 0$$

## 2.4 Derivatives of trigonometric functions

Trigonometric limits:

$$1. \lim_{\theta \rightarrow 0} \frac{\sin \theta}{\theta} = 1$$

$$2. \lim_{\theta \rightarrow 0} \frac{\cos \theta - 1}{\theta} = 0$$

Trigonometric derivatives,

$$1. \frac{d}{dx} \sin x = \cos x$$

$$2. \frac{d}{dx} \cos x = -\sin x$$

$$3. \frac{d}{dx} \tan x = \sec^2 x$$

$$4. \frac{d}{dx} \csc x = -\csc x \cot x$$

$$5. \frac{d}{dx} \sec x = \sec x \tan x$$

$$6. \frac{d}{dx} \cot x = -\csc^2 x$$

## 2.5 Chain Rule

If  $g$  is differentiable at  $x$  and  $f$  is differentiable at  $g(x)$ ,  $F = f \circ g$  i.e.  $F(x) = f(g(x))$

$$F'(x) = f'(g(x)) \cdot g'(x)$$

Alternatively, if  $y = f(u)$  and  $u = g(x)$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}, \text{ Leibniz}$$

## 3.1 Maximum and minimum.

### Definition of max and min

If  $c$  is a number in the domain  $D$  of function  $f$ ,  $f(c)$  is a

abs. max if  $f(c) \geq f(x) \quad \forall x \in D$

abs min if  $f(c) \leq f(x) \quad \forall x \in D$

local max if  $f(c) \geq f(x)$  when  $x$  is near  $c$

local min if  $f(c) \leq f(x)$  when  $x$  is near  $c$ .

### ★ Extreme value theorem, EVT

Given  $f$  is continuous on a closed interval  $[a, b]$ ,

$f$  attains an abs max value  $f(c)$  and abs min value  $f(d)$  at some numbers  $c$  and  $d$  in  $[a, b]$ .

### Boundedness theorem for continuous functions

Given  $f$  is defined and continuous on the closed interval  $[a, b]$ ,

$f$  is bounded on  $[a, b]$

### Definition of critical numbers:

A critical number of function  $f$  is a number  $c$  in domain of  $f$  st.

$f'(c) = 0$  or  $f'(c)$  DNE

### Fermat's theorem

If  $f$  has a local max/min at  $c$ , and  $f'(c)$  exists

$f'(c) = 0$ , Note, vice versa is not necessarily true

→ i.e. if  $f$  has a local min/max at  $c$ ,  $c$  is a critical number of  $f$ .

The Closed Interval Method to find abs max/min.

1. Find  $f$  @ end points,  $a$  and  $b$ .
2. Find  $f$  @ crit numbers in  $[a, b]$
3. Find the largest and smallest numbers

3.2 Mean Value theorem.

Rolle's theorem

- Given
1.  $f$  is continuous on the closed interval  $[a, b]$
  2.  $f$  is differentiable on open interval  $(a, b)$
  3.  $f(a) = f(b)$ ,

There must be a number  $c \in (a, b)$  s.t.  $f'(c) = 0$

Mean value theorem (MVT)

- Given
1.  $f$  is continuous on the closed interval  $[a, b]$
  2.  $f$  is differentiable on open interval  $(a, b)$

There must be a number  $c \in (a, b)$  s.t.  $f'(c) = \frac{f(b) - f(a)}{b - a}$  or  $f(b) - f(a) = f'(c)(b - a)$

Results from MVT:

If  $f'(x) = 0 \quad \forall x \in (a, b)$ ,  $f$  is constant on  $(a, b)$

If  $f'(x) = g'(x) \quad \forall x \in (a, b)$ , then  $f - g$  is constant  $(a, b)$   
 $f(x) = g(x) + c$ ,  $c$  is a constant

3.3 How Derivatives affect the shape of a graph

Increasing/Decreasing test (QT1)

If  $f'(x) > 0$  on an interval, then  $f$  is increasing on that interval.

If  $f'(x) < 0$  on an interval, then  $f$  is decreasing on that interval.

First derivative test (QT2)

Given  $c$  is a critical number of continuous function  $f$ .

1. if  $f$  changes from +ve to -ve at  $c$ ,  $f$  has a local max at  $c$ .
2. if  $f$  changes from -ve to +ve at  $c$ ,  $f$  has a local min at  $c$ .
3. if  $f$  does not change signs at  $c$ ,  $f$  has no local min/max at  $c$ .

Definition of concavity

If  $f$  lies above all of its tangents on an interval  $I$ , then it is concave upward on  $I$ .

If  $f$  lies below all of its tangents on an interval  $I$ , then it is concave downward on  $I$ .

### Concavity Test (Q3)

- a. If  $f''(x) > 0 \forall x$  in  $I$ ,  $f$  is concave up on  $I$   
b. If  $f''(x) < 0 \forall x$  in  $I$ ,  $f$  is concave down on  $I$

### Second Derivative Test (Q4)

Given  $f''$  is continuous near  $c$

- a. If  $f'(c) = 0$  and  $f''(c) > 0$ , then  $f$  has a local minimum at  $c$ .  
b. If  $f'(c) = 0$  and  $f''(c) < 0$ , then  $f$  has a local maximum at  $c$ .

## 3.4. Horizontal asymptotes

Definition of infinite limits:

Given  $f$  is a function defined on some interval  $(a, \infty)$ , then

$\lim_{x \rightarrow \infty} f(x) = L$  means that  $f(x)$  is arbitrarily close to  $L$  by taking  $x$  sufficiently large

Similarly  $x \rightarrow -\infty$  represents the same.

Definition of horizontal asymptotes:

The line  $y = L$  is a horizontal asymptote of  $y = f(x)$  if either

$$\lim_{x \rightarrow -\infty} f(x) = L \quad \text{or} \quad \lim_{x \rightarrow \infty} f(x) = L$$

Some theorem  $\boxed{-}$ :

if  $r > 0$

$$\lim_{x \rightarrow \infty} \frac{1}{x^r} = 0 \quad \text{and} \quad \lim_{x \rightarrow -\infty} \frac{1}{x^r} = 0$$

Rigorous definition of infinite limits.

$$\lim_{x \rightarrow \infty} f(x) = L \quad \text{means that} \quad \forall \epsilon > 0, \exists N \text{ s.t. } x > N \Rightarrow |f(x) - L| < \epsilon$$

$$\lim_{x \rightarrow -\infty} f(x) = L \quad \text{means that} \quad \forall \epsilon > 0, \exists N \text{ s.t. } x < -N \Rightarrow |f(x) - L| < \epsilon$$

$$\lim_{x \rightarrow \infty} f(x) = \infty \quad \text{means that} \quad \forall M > 0, \exists N > 0 \text{ s.t. } x > N \Rightarrow f(x) > M$$

And related limits with  $\pm \infty$

### 3.5 Curve sketching

See curve sketching outline.

### 2.9 Linear approximation and differentials

Linear approximation:  $f(x) \approx f(a) + f'(a)(x-a)$

### 3.9 Antiderivatives

$F(x)$  is an antiderivative of  $f(x)$  if  $F'(x) = f(x)$

### 4.2 Definite Integral

Definition of definite integrals

$$\int_a^b f(x) dx \equiv \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i) \Delta x, \quad \Delta x = \frac{(b-a)}{n}, \quad x_i = a + i \Delta x \leftarrow \text{generally}$$

$\hookrightarrow$  Note  $x_i$  can be any # b/w and including  $[x_{i-1}, x_i]$   
left endpoint right endpoint

Definition of the Riemann Sum:

$$R_n = \sum_{i=1}^n f(x_i) \Delta x$$

$\hookrightarrow$  Note  $R_n$  is an approximation of the integral

$$\text{also, } \int_a^b f \equiv \lim_{n \rightarrow \infty} R_n$$

Sum of numbers:

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}$$

$$\sum_{i=1}^n i^2 = \frac{n(2n+1)(n+1)}{6}$$

$$\sum_{i=1}^n i^3 = \left[ \frac{n(n+1)}{2} \right]^2$$

Midpoint Rule:

$$\begin{aligned} \int_a^b f(x) dx &\approx \sum_{i=1}^n f(\bar{x}_i) \Delta x \\ &= \Delta x [f(\bar{x}_1) + f(\bar{x}_2) + \dots + f(\bar{x}_n)] \end{aligned}$$

$$\Delta x = \frac{b-a}{n} \quad \text{and} \quad \bar{x}_i = \frac{1}{2}(x_{i-1} + x_i), \text{ middle of } [x_{i-1}, x_i]$$

## Properties of the integral

Equal-wise:

$$\rightarrow \int_a^b f(x) dx = - \int_b^a f(x) dx$$

$$\rightarrow \int_a^a f(x) dx = 0$$

$$\rightarrow \int_a^b c dx = c(b-a), c \text{ is any constant}$$

$$\rightarrow \int_a^b f(x) + g(x) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

$$\rightarrow \int_a^b c f(x) dx = c \int_a^b f(x) dx, c \text{ is any constant}$$

$$\rightarrow \int_a^b [f(x) - g(x)] dx = \int_a^b f(x) dx - \int_a^b g(x) dx$$

$$\rightarrow \int_a^b f(x) dx = \int_a^c f(x) dx + \int_c^b f(x) dx$$

Comparison-wise:

$$\rightarrow \text{If } f(x) \geq 0 \text{ for } x \in [a, b], \\ \int_a^b f(x) dx \geq 0$$

$$\rightarrow \text{If } f(x) \geq g(x) \text{ for } x \in [a, b], \\ \int_a^b f(x) dx \geq \int_a^b g(x) dx$$

$$\rightarrow \text{If } m \leq f(x) \leq M \text{ for } x \in [a, b], \\ m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

$$\rightarrow \left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$$

## 4.3 Fundamental Theorem of Calculus

Part I:

Given: If  $f$  is cont on  $[a, b]$ :

$$\text{Then: } g(x) = \int_a^x f(t) dt \quad a \leq x \leq b$$

is cont on  $[a, b]$  and diff on  $(a, b)$

$$g'(x) = f(x)$$

Part II:

Given: If  $f$  is cont on  $[a, b]$

$$\text{Then: } \int_a^b f(x) dx = F(b) - F(a), \text{ where } F' = f$$

## 4.4 Indefinite integrals and net change theorem

Net change Theorem:

↳ The integral of rate of change is the net change

$$\int_a^b f'(x) dx = F(b) - F(a), \text{ where } f'(x) \text{ is the rate of change,} \\ F(b) - F(a) \text{ is the change in } y \text{ as } x \text{ moves from } a \text{ to } b$$

or simply:

$$\int_a^b \frac{dF}{dx} dx$$

$$= \int_a^b dF$$

$$= \Sigma \Delta F$$

$$= \Delta F$$

## 4.5 Substitution Rule

Substitution Rule (or "u-sub"):

Given  $u = g(x)$  is diff whose range is an interval and is cont on  $I$ , then

$$\int f(g(x))g'(x) dx$$

$$= \int f(u) du$$

Substitution rule for definite integrals:

Given:  $g'$  is cont. on  $[a, b]$  and  $f$  is cont on the range of  $u = g(x)$

$$\text{Then: } \int_a^b f(g(x))g'(x) dx = \int_{g(a)}^{g(b)} f(u) du$$

Symmetry and integrals

① If  $f$  is cont on  $[-a, a]$  and  $f$  is even,

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

② If  $f$  is cont on  $[-a, a]$  and  $f$  is odd,

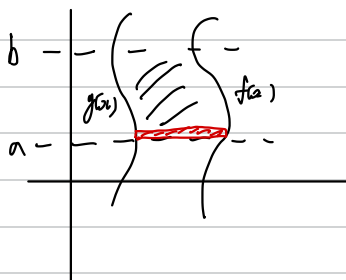
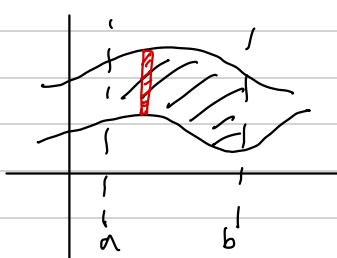
$$\int_{-a}^a f(x) dx = 0$$

5.1 Area b/w curves

$$\text{Area} = \int_a^b [f(x) - g(x)] dx$$

where  $f(x) \geq g(x)$

↳ Same can go for y  
direction



Generally,

$$\text{Area} = \int_a^b |f(x) - g(x)| dx$$

## 5.2 Volume

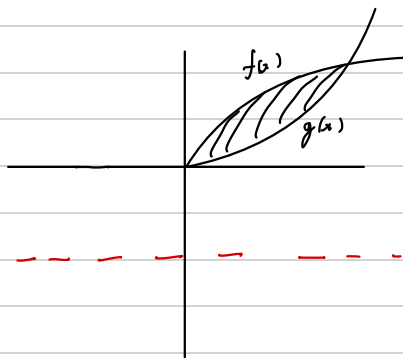
Def'n of volume:

$$\hookrightarrow V = \int_a^b f(x) dx$$

Volume by washer/disk method:

$$V = \int_a^b A(x) dx, \quad A(x) = \pi \left[ \overbrace{f(x)-k}^{\text{outer radius}^2} - \overbrace{g(x)-k}^{\text{inner radius}^2} \right]$$

$$V = \pi \int_a^b [f(x)-k]^2 - [g(x)-k]^2 dx$$



Note: function variable is the same as the axis of rotation

## 5.3 Volumes by cylindrical shells

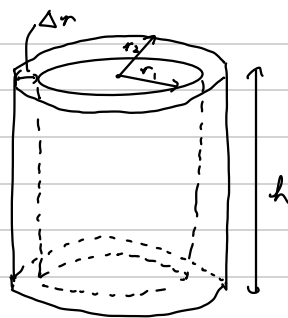
- In general, volume of a cylindrical shell:

$$V = 2\pi \left( \frac{r_1 + r_2}{2} \right) \cdot h (r_2 - r_1)$$

If  $r = \frac{r_1 + r_2}{2}$ , average radius,  $\Delta r = r_2 - r_1$ , change in radius

$$= 2\pi r h \Delta r$$

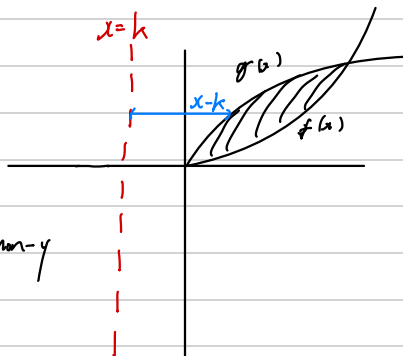
$$= (\text{circumference}) (\text{height}) (\text{thickness})$$



Volume by cylindrical shells

$$V = \int_a^b 2\pi x f(x) dx$$

for area b/w curve and non-y rotation



$$V = \int_a^b 2\pi \underbrace{(x-k)}_{\text{radius}} \underbrace{(f(x)-g(x))}_{\text{height}} dx$$

## 5.5 Average value of f(x)

Average value is:

$$f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx$$

Second MVT for integrals:

Given:  $f, g$  are cont on  $[a, b]$   
 $\Rightarrow g$  is +ve

MVT for integrals:

Given:  $f(x)$  is cont on  $[a, b]$ ,

Then: There exists a  $c \in [a, b]$  s.t.

Then: There exists a  $c \in [a, b]$  s.t.

$$\int_a^b f(x) g(x) dx = f(c) \int_a^b g(x) dx$$

$$f(c) = f_{\text{avg}} = \frac{1}{b-a} \int_a^b f(x) dx \quad \text{or} \quad \int_a^b f(x) dx = f(c)(b-a)$$

## 6.1 Inverse Functions

Definition of one-to-one function:

$$f(x_1) \neq f(x_2) \text{ whenever } x_1 \neq x_2$$

↳ Do a horizontal line test

Theorems with 1-1 functions:

- If  $f$  is inc or dec, then  $f$  is 1-1

Definition of inverse function:

- If  $f$  is a one-to-one function with domain  $A$  and range  $B$ .

Then the inverse function  $f^{-1}$

↳ has domain  $B$  and range  $A$

↳ defined as  $f^{-1}(y) = x \iff f(x) = y$

Cancellation equations:

$$f^{-1}(f(x)) = x \text{ for every } x \text{ in } A$$

$$f(f^{-1}(y)) = y \text{ for every } y \text{ in } B$$

Continuity of inverse function theorem:

- If  $f$  is an one-to-one function defined on an interval, then its inverse  $f^{-1}$  is also cont.

Derivative of  $f^{-1}(a)$

If  $f$  is a one-to-one diff  $f$  with inverse  $f^{-1}$

and  $f'(f^{-1}(a)) \neq 0$ ,

$$f^{-1} \text{ is diff at } a, \quad f^{-1}(a) = \frac{1}{f'(f^{-1}(a))}$$

In Leibniz,

$$\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$$

$f^{-1}(x)$  is the reflection of  $f(x)$  on  $y=x$

## 6.2 Natural Logarithmic Function

Def'n of  $\ln x$ :

$$\ln x = \int_1^x \frac{1}{t} dt \quad x > 0$$

If  $0 < x < 1$ ,

$$\ln x = - \int_x^1 \frac{1}{t} dt$$

Definition of logarithmic functions:

Def'n: A logarithmic function is a non-constant differentiable function  $f$ , defined for  $x \in \{\mathbb{R}, (0, \infty)\}$  such that for all  $a > 0$  and  $b > 0$ :

$$f(a \cdot b) = f(a) + f(b)$$

Properties of logarithms:

$$(1) f(1) = 0$$

$$(2) f\left(\frac{1}{x}\right) = -f(x)$$

$$(3) f\left(\frac{x}{y}\right) = f(x) - f(y)$$

$$(4) f'(x) = \frac{1}{x} f'(1)$$

Properties of  $\ln x$ :

$$(1) \ln(x) = \frac{1}{x} > 0, \ln x \text{ is increasing}$$

$$(9) \ln(x) = \log_e x$$

$$(2) \ln(x) \text{ is cont \& diff}$$

$$(10) (\ln x)'' = -\frac{1}{x^2} < 0, \text{C.D.}$$

$$(3) x > 1 \text{ for } \ln(x) > 0$$

$$x < 1 \text{ for } \ln(x) < 0$$

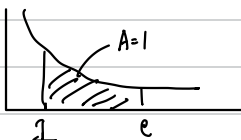
$$(4) \ln(a \cdot b) = \ln(a) + \ln(b)$$

$$(5) \ln x^{\frac{p}{q}} \\ = \frac{p}{q} \ln x$$

$$(6) \text{Range of } \ln(x) \text{ is } (-\infty, \infty)$$

$$(7) \lim_{x \rightarrow \infty} \ln x = \infty \\ \lim_{x \rightarrow 0^+} \ln x = -\infty$$

$$(8) \ln(e) = 1 \rightarrow$$



### 6.3 Natural exponential function

$$\rightarrow y = e^x \text{ s.t. } \ln e^x = x \text{ or } e^{\ln x} = x \quad (x > 0)$$

Limits of  $e^x$ :

$$\lim_{x \rightarrow -\infty} e^x = 0 \quad \lim_{x \rightarrow +\infty} e^x = \infty$$

Properties of  $e^x$ :

$$1. \ln e^x = x \text{ or } e^{\ln x} = x, x > 0$$

$$2. e^x > 0$$

$$3. e^0 = 1$$

$$4. e^{a+b} = e^a \cdot e^b$$

$$5. e^{-c} = \frac{1}{e^c}, \quad e^{a-b} = \frac{e^a}{e^b}$$

$$6. \frac{d}{dx} e^x = e^x$$

$$7. \int e^x dx = e^x + C$$

$$\rightarrow e^x > 1+x$$

$$\hookrightarrow e^x = 1 + \int_0^x e^t dt$$

$$\text{Given } e^x > 1,$$

$$e^x > 1$$

$$1 + \int_0^x e^t dt > 1 + \int_0^x 1 dt$$

$$\underline{\underline{e^x > 1+x}}$$

## 6.4 General logarithmic and exponential function

$$b^x = e^{\ln b^x} \\ = e^{x \ln b}$$

Properties:

$$\rightarrow b^{x+y} = b^x (b^y)$$

$$\rightarrow b^{x-y} = \frac{b^x}{b^y}$$

$$\rightarrow (b^x)^y = b^{xy}$$

$$\rightarrow (ab)^x = a^x b^x$$

$$\rightarrow \frac{d}{dx} b^x = (\ln b) b^x$$

$$\rightarrow \int b^x dx = \frac{b^x}{\ln b} + C$$

$\rightarrow$  Change of base formula:

$$\log_b x = \frac{\ln x}{\ln b}$$

$$\rightarrow \frac{d}{dx} \log_b x = \frac{1}{x \ln b}$$

$$e = \lim_{x \rightarrow 0} (1+x)^{\frac{1}{x}} \quad , e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

## 6.6 Inverse trigonometric functions

Derivative of inverse trig functions:

$$\frac{d}{dx} \sin^{-1} x = \frac{1}{\sqrt{1-x^2}} \quad , \quad \frac{d}{dx} \cos^{-1} x = \frac{-1}{\sqrt{1-x^2}} \quad , \quad \frac{d}{dx} \tan^{-1} x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \csc^{-1} x = \frac{-1}{x \sqrt{x^2-1}} \quad , \quad \frac{d}{dx} \sec^{-1} x = \frac{1}{x \sqrt{x^2-1}} \quad , \quad \frac{d}{dx} \cot^{-1} x = \frac{-1}{1+x^2}$$

Integrals with inverse trig functions:

$$\int \frac{1}{x^2 + a^2} dx \quad \int \frac{dx}{\sqrt{a^2 - x^2}} \quad \int \frac{dx}{x \sqrt{x^2 - a^2}}$$

$$= \frac{1}{a} \tan^{-1} \left( \frac{x}{a} \right) + C \quad = \sin^{-1} \left( \frac{x}{a} \right) + C \quad = \frac{1}{a} \sec^{-1} \left( \frac{|x|}{a} \right) + C$$

## 6.8 L'Hospital's Rule

### L'Hospital's Rule.

Given:  $f, g$  are diff and  $g'(x) \neq 0$

$$\left. \begin{array}{l} \text{and } \lim_{x \rightarrow a} f(x) = 0 \text{ and } \lim_{x \rightarrow a} g(x) = 0 \\ \text{or } \lim_{x \rightarrow \infty} f(x) = \pm \infty \text{ and } \lim_{x \rightarrow \infty} g(x) = \pm \infty \end{array} \right\} \frac{0}{0} \text{ or } \frac{\infty}{\infty}$$

$a = \text{constant}, 0, \pm \infty$  (left side or right side)

$$\begin{aligned} \text{Then: } \lim_{x \rightarrow a} \frac{f(x)}{g(x)} &= \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} \\ &= L \text{ or } \pm \infty \end{aligned}$$

### Indeterminate forms (and ways to approach)

$$\begin{array}{l} 0 \cdot \infty: \lim_{x \rightarrow a} f(x) = 0 \quad \text{Rewrite it as} \\ \lim_{x \rightarrow a} g(x) = \infty \quad \lim_{x \rightarrow a} \frac{f(x)}{\frac{1}{g(x)}} \text{ and apply LH} \end{array}$$

Indeterminate  $\lim_{x \rightarrow a} f(x)g(x) = 0 \cdot \infty$

$$\begin{array}{l} \infty - \infty: \lim_{x \rightarrow a} f(x) = \infty \quad \text{factor and rewrite as quotient} \\ \lim_{x \rightarrow a} g(x) = \infty \quad \text{and apply LH} \end{array}$$

Indeterminate  $\lim_{x \rightarrow a} f(x) - g(x) = \infty - \infty$

$$\begin{array}{l} 0^0, \infty^0, 1^\infty: \lim_{x \rightarrow a} [f(x)]^{g(x)} = 0^0 \\ \text{or } \infty^0 \\ \text{or } 1^\infty \end{array} \quad \begin{array}{l} \text{Rewrite as:} \\ f(x)^{g(x)} = e^{g(x) \ln f(x)} \\ \lim_{x \rightarrow a} f(x)^{g(x)} = e^{\left[ \lim_{x \rightarrow a} g(x) \ln f(x) \right]} \end{array}$$

Indeterminate

### Cauchy's Mean Value Theorem:

Given:  $f$  and  $g$  are cont. on  $[a, b]$  and diff. on  $(a, b)$   
 $g'(x) \neq 0$  for all  $x$  in  $(a, b)$

Then: There exists a  $c$  in  $(a, b)$  s.t.

$$\frac{f'(c)}{g'(c)} = \frac{f(b) - f(a)}{g(b) - g(a)}$$

# Differential equations 9.1-9.6, 6.5

## Separable equations

$$\frac{dy}{dx} = \frac{g(x)}{h(y)}$$

$$\int h(y) dy = \int g(x) dx$$

## Linear diff eq.

$$y' + p(x)y = q(x)$$

$$IF = e^{\int p(x) dx}$$

## Newton's Law of cooling

$$\frac{dT}{dt} = k(T - T_s)$$

$$T(t) = e^{kt+C} + T_s \\ = e^C(e^{kt}) + T_s$$

## Population Growth

↳ Natural growth

$$\frac{dP}{dt} = kP \Rightarrow P(t) = P_0 e^{kt}$$

↳ Logistic differential equation

$$\frac{dP}{dt} = kP \left(1 - \frac{P}{M}\right)$$

$$P(t) = \frac{M}{1 + A e^{-kt}}, \quad A = \frac{M - P_0}{P_0}$$

## Interest

$$\hookrightarrow A_0 \left(1 + \frac{r}{n}\right)^{nt}$$

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

or

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^{\frac{1}{n}}$$

## Orthogonal trajectory

Steps:

1. find  $\frac{dy}{dx}$
2. sub for  $k$
3. Reciprocal
4. Solve w sep eq.

## RL Circuit

$$V = RI$$

$$V = L \frac{dI}{dt}$$

## Complex numbers

$$\hookrightarrow i^2 = -1$$

modulus,  $|z|$

$$\hookrightarrow z = a + ib$$

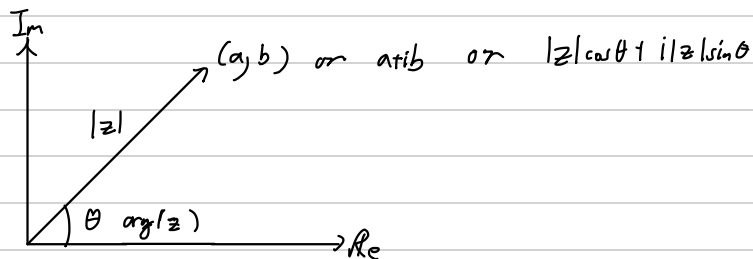
$$= |a + ib|$$

$$\operatorname{Re}(z) = a$$

$$= \sqrt{a^2 + b^2}$$

$$\operatorname{Im}(z) = b$$

argument,  $\arg(z)$



$$\text{Polar: } z = r \cos \theta + i r \sin \theta$$

$$|z_1 \cdot z_2| = |z_1| |z_2|$$

$$|z| \cos(\arg z) = a$$

$$\arg(z_1 z_2) = \arg(z_1) + \arg(z_2)$$

$$|z| \sin(\arg z) = b$$

$$\tan(\arg z) = b/a$$

$$z_1 z_2 = |z_1| |z_2| (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2))$$

$$z^n = r^n (\cos n\theta + i \sin n\theta)$$

## De Moivre's Theorem:

$$e^{ix} = \cos x + i \sin x$$

## 17.1 Homo, 2<sup>nd</sup> order, const. coeff. diff eq.

Solution to

$$ay'' + by' + cy = 0$$

Aux eq.  $= ar^2 + br + c = 0$

If  $b^2 - 4ac > 0$ ,

If  $b^2 - 4ac = 0$

If  $b^2 - 4ac < 0$

Solution is

$$y = C_1 e^{r_1 x} + C_2 e^{r_2 x}$$

Solution is

$$y = C_1 e^{rx} + C_2 x e^{rx}$$

Solution is

$$y = e^{\alpha x} (C_1 \sin \beta x + C_2 \cos \beta x)$$

## 17.2 Guessing game

Solution:

$$y(x) = y_p(x) + y_c(x)$$

Undeterm

$$u_1' y_1 + u_2' y_2 = 0$$